

SLOPEWATCH · DESIGN DECISIONS REGISTER

SlopeWatch design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	SLW-DEC-001	Version	0.1
Status	Draft	Date	2026-10-01
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/slopedwatch		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, SLW-DDR-001 to SLW-DDR-003 and the build plan work; budget treated as a value-engineering target

SlopeWatch design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design-for-construction changes P1 to P11	Accept as made; ask for changes	Accept	Every component of the build plan	SLW-DDR-003, Table 1
2	Mast margin in wind with the side-mounted alert unit (factor 1.7 on yield in a 35 m/s gust, was 2.1)	(a) accept for the prototype; (b) narrow the alert back plate below the box; (c) a 60.3 mm mast	(a), with a check against local wind data before a site installation	Mast (site option), alert back plate	SLW-DDR-003, A1; SLW-CAL-001 [H2]
3	Mounting on larger poles or walls: the V-block fixing fits round poles of 40 to 70 mm, while the reference site says "an existing pole or building"	(a) state the reference site as a 40 to 70 mm pole, the mast otherwise; (b) add four wall holes to both back plates for coach screws; (c) longer bands and a wider V	(b)	Alert unit back plate; FieldNode back plate (cross-repo)	SLW-DDR-003, A2
4	Pole for the first prototype	(a) build the mast option; (b) an existing 48 mm pole at the test site	(a)	Steps 13 to 16	SLW-DDR-003, A3
5	The <code>problem</code> wording in <code>project.yaml</code> ("often give warning"), changed at TRL 2	Accept; revert	None made	None	SLW-DDR-001 O1, SLW-DDR-002
6	First co-design partner and site type	Artisanal mining cooperative; quarry; hillside community with a district disaster office	None made	Test slope for steps 2 to 19	SLW-DDR-001 O2
7	Crack-gauge thresholds (placeholders 1 mm per day over 24 h, 1 mm per hour)	Set per site with a geotechnical partner	None made	Firmware alert rules, not the hardware	SLW-DDR-001 O3
8	Alarm reach near running machinery (R7 not met there)	Second alert unit at the work face, linked by LoRa; radio pager for operators	None made	A second alert unit would repeat sections 3.11 to 3.13	SLW-PRC-001, open questions; SLW-CAL-001 [E3]
9	Protecting the surface cable from rockfall and movement, and detecting a cut cable	Bury where the ground allows; armoured cable; a fault message when a stake stops answering	None made	Bus conduit (step 19); firmware	SLW-PRC-001, open questions
10	Stake head colour and marking (appearance session, 2026-09-26, items 1 and 2)	Signal amber with a retroreflective band; stake ID label and a downslope arrow on the crown	Adopt both	Stake head paint and label (section 3.5)	REVIEW 2026-09-26
11	FieldNode 12 V rail rating of at least 0.5 A continuous (R8), requested of FieldNode	Wait for FieldNode; give the alert unit its own small battery	Wait for FieldNode (decided as a request on 2026-09-25)	Alert unit power; first	SLW-DDR-002 item 9

#	Decision needed	Options	Recommendation	Affects in the build	Source
	under SLW-DDR-002 and not yet rated there			check "Alert rail current"	
12	Sensor port pin assignment of the FieldNode M12 ports (needed for the bus plug and the alert plug, including the switch input on port B)	Agreed with FieldNode	None made	Field-wired M12 plugs (section 3.17)	FND-DDR-001 O2

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The 1.5 in end cap thread matches the pipe thread (BSP or NPT)	The cap must seal against grout and water	SLW-DDR-003, P2
2	The 110 x 50 mm reducer's 50 mm socket is 50 to 51 mm inside	The tape wrap takes up only about 1 mm a side	SLW-DDR-003, P3
3	The capsule board is at most 28 mm wide and fits the 30 mm bore of the tube	Sets the capsule tube size	SLW-DDR-003, P1
4	The displacement sensor has an M6 rod-end ball joint at each end and a removable plunger-end joint, and its plunger thread fits the coupling	The rod and the ball joints are sized to it	SLW-DDR-003, P5
5	The alert and switch boxes have four corner screw holes outside the lid seal, and their spacing matches the plate holes	The plates are drilled to suit	SLW-DDR-003, P8, P9
6	The horn has a sealing flange and the beacon a gasketed base that cover a 10 mm hole	Keeps the alert box sealed	SLW-DDR-003, P8
7	The conduit fittings' hole size and lock nut size fit inside the 110 mm head	Sets the 20.5 mm holes and the lock nut clearance	SLW-DDR-003, P4
8	The existing pole at the test site, if used, is round, 40 to 70 mm, and sound enough to carry the node and alert unit in wind	The fixing and the wind case depend on it	SLW-CAL-001 [H1]

■ Value engineering

Value-engineering target: USD 250 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 296.50 for the SlopeWatch-specific parts of the reference site, node on an existing pole (USD 46.50 over the target); USD 324.50 where the optional mast is needed. The FieldNode core (USD 126) is costed in its own project. Main cost drivers and savings worth trying:

- The largest lines are the three tilt sensor capsules (USD 82.50, mostly the three inclinometers at about USD 18 each), the crack gauge with its reader (USD 46), the alert unit with its switch post (USD 43), the bus cable in conduit (USD 42) and the three stakes (USD 39).
- Making the design constructable added USD 51.50: end caps and stand tubes (USD 12), collars (USD 1.50), stake heads from drainage fittings with conduit fittings and connectors (USD 18), the gauge's clamp blocks, ball

joints, rod coupling, folded guard and pegs (USD 8), the alert and switch mounting plates and clamps (USD 10), and tape and a plug (USD 2).

- Savings worth trying: bury the bus cable without conduit where the ground is soft and stable (up to about USD 15); buy the inclinometers and boards in a batch for several sites; make the guard from offcut sheet; use plain M20 cable glands on the stake heads where the conduit stops short of the head (about USD 2 a stake); price a second-source crack sensor with ball joints included.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: FieldNode core costed in FieldNode; tailings dams kept in the pitch with limits; surface tilt stakes plus a crack gauge; wired RS-485 bus; SCL3300 in mode 1; published tilt-rate thresholds and keyed 30 min silence; alert unit at the toe	Amish: "i accept all your recommendations, go with them across all repos."	SLW-DDR-001, SLW-DDR-002
2026-09-25	Reference site on an existing pole, mast a site option; FieldNode asked to rate its 12 V rail; keyed switch on its own post about 5 m from the siren; precaution beacon at 1 % duty or less; own TwinKit gateway at SF12 sites	Amish, same instruction: go with recommendation	SLW-DDR-002, items 8 to 12
2026-09-30	Make the design physically buildable while drawing the build plan, keeping what the product does; changes recorded for review	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SLW-DDR-003 (changes open for his review)
2026-09-30	Build plan format approved; open decisions kept out of the build plan, in this register	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"	SLW-BLD-001, SLW-DEC-001
2026-10-01	<code>budget_usd</code> is a hypothetical value-engineering target, not a limit; cost reported against it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register, Value engineering; SLW-CAL-001 v0.3